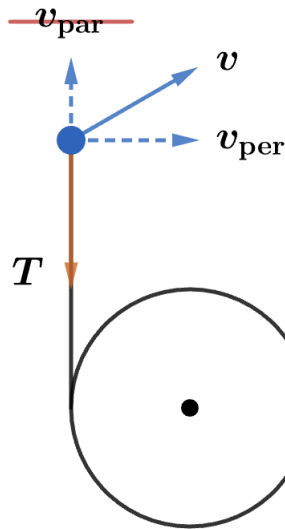
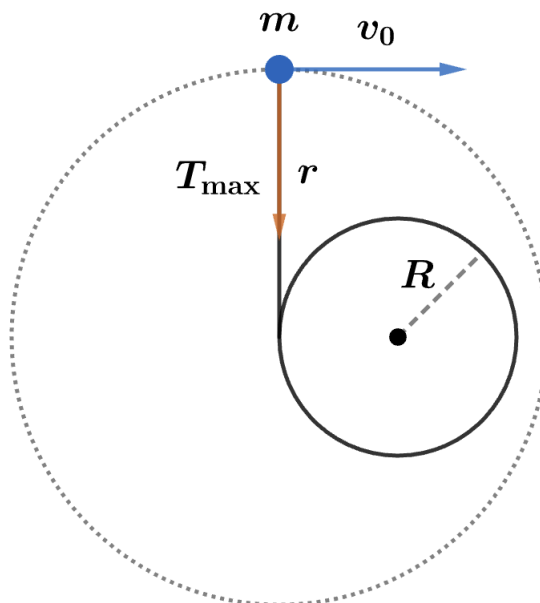


2007 F=ma Exam: Problem 34

Kevin S. Huang



Note that the velocity is always perpendicular to the tension force by conservation of string, since having a parallel component would mean the total string length is changing (not just the unwound portion). Because perpendicular forces do no work, the object has a constant kinetic energy and hence constant speed $v = v_0$.



The angular momentum L of the object when the rope breaks is given by

$$L = mv_0 r$$

where r is the length of the unwound rope at this instant. To relate r to $T_{\max}$, note that the object approximately moves in a circle of radius r when the rope breaks so

$$T_{\max} = \frac{mv_0^2}{r}$$

$$r = \frac{mv_0^2}{T_{\max}}$$

Thus,

$$L = mv_0 \left(\frac{mv_0^2}{T_{\max}} \right) = \frac{m^2 v_0^3}{T_{\max}}$$

so the answer is B.